

Higher order linear differential equations

The general form of n^{th} order linear diff. eq is given by

$$P_0(x) \frac{d^n y}{dx^n} + P_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1}(x) \frac{dy}{dx} + P_n(x) y = \phi(x)$$

where $P_0(x), P_1(x), \dots, P_n(x)$ and $\phi(x)$ are functions of x

Linear diff. eq. with constant coefficients

A diff. eq. of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x) \rightarrow (I)$$

where $a_0, a_1, a_2, \dots, a_n$ are constants, is the most general form of a linear diff. eq with const. coeffs.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x) \rightarrow (II) \text{ is the general}$$

form of a second order linear diff. eq with const coeffs.

Associated diff. eq. of eq (II) is given by

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0 \rightarrow (III)$$

It is also called homogeneous part of eq (II)

Solution of second order linear homogeneous equation

Consider the diff eq.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$$

Dividing by a_0 we get

$$\frac{d^2 y}{dx^2} + \frac{a_1}{a_0} \frac{dy}{dx} + \frac{a_2}{a_0} y = 0$$

$$\text{or } \frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = 0 \rightarrow (I) \quad (P = \frac{a_1}{a_0}, Q = \frac{a_2}{a_0})$$

since this is a diff eq with const coeff
we assume a trial solution $y = e^{mx}$

$$\frac{dy}{dx} = m e^{mx} \quad \frac{d^2y}{dx^2} = m^2 e^{mx}$$

Sub in (1)

$$m^2 e^{mx} + p m e^{mx} + q e^{mx} = 0$$

$$e^{mx} (m^2 + pm + q) = 0$$

This is true only if

$$m^2 + pm + q = 0 \rightarrow (2) \quad (\because e^{mx} \neq 0)$$

This is quadratic eq in m . It has two roots. $\Delta = p^2 - 4q$ is the discriminant

Case (I) $p^2 - 4q > 0$. eq (2) has two distinct real roots say m_1 and m_2 . $\therefore e^{m_1 x}$ and $e^{m_2 x}$ are independent solutions

$\therefore y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ is the gen. soln.

Case (II) $p^2 - 4q = 0$ eq (2) has two equal roots. i.e. $m_2 = m_1$. In this case we get two independent solutions $e^{m_1 x}$ and $x e^{m_1 x}$

$\therefore$ General solution of (2) is given by

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x} = (c_1 + c_2 x) e^{m_1 x}$$

Case (III) $p^2 - 4q < 0$. eq (2) has two complex roots say $\alpha \pm i\beta$. The gen. soln is given

General solution is given by

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x}$$

$$= c_1 e^{\alpha x} \cdot e^{i\beta x} + c_2 e^{\alpha x} \cdot e^{-i\beta x}$$

$$= e^{\alpha x} [c_1 e^{i\beta x} + c_2 e^{-i\beta x}]$$

$$= e^{\alpha x} [c_1 (\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)]$$

$$\bar{w} \quad y = e^{\alpha x} [(c_1 + c_2) \cos \beta x + i(c_1 - c_2) \sin \beta x]$$

$$\bar{w} \quad y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x]$$

$$\text{where } C_1 = c_1 + c_2 \quad C_2 = i(c_1 - c_2)$$

Solve the following diff. eqs

$$(1) \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

$$(2) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$$

$$(3) \frac{d^2 y}{dx^2} + 4y = 0$$

$$(4) \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + 29y = 0$$

$$(1) \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$

$$\text{Let } D \equiv \frac{d}{dx} \text{ and } D^2 \equiv \frac{d^2}{dx^2}$$

$$\therefore \text{ we get } D^2 y + 5Dy + 6y = 0 \Rightarrow (D^2 + 5D + 6)y = 0$$

$$\bar{w} \quad D^2 + 5D + 6 = 0 \text{ is the Auxillary eq.}$$

$$\bar{w} \quad (D+2)(D+3) = 0 \Rightarrow D = -2, -3$$

General solution is given by

$$y = C_1 e^{-2x} + C_2 e^{-3x}$$

$$(2) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad \bar{w} \quad D^2 y - 2Dy + y = 0$$

$$\Rightarrow (D^2 - 2D + 1)y = 0$$

Auxillary eq is given by

$$D^2 - 2D + 1 = 0 \Rightarrow (D-1)^2 = 0 \quad D = 1, 1$$

$$\text{General soln, } y = (C_1 + C_2 x) \cdot e^x$$

$$(3) \frac{d^2 y}{dx^2} + 4y = 0 \Rightarrow (D^2 + 4)y = 0$$

$$\text{Auxillary eq is } D^2 + 4 = 0 \Rightarrow D^2 = -4 \Rightarrow D = \pm 2i$$

$$\bar{w} \quad D = 0 \pm 2i \quad (\alpha = 0, \beta = 2)$$

General solution

$$\begin{aligned} y &= e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] \\ &= e^{0x} [C_1 \cos 2x + C_2 \sin 2x] \\ &= C_1 \cos 2x + C_2 \sin 2x \end{aligned}$$

$$(4) \frac{d^2 y}{dx^2} + 10 \frac{dy}{dx} + 29y = 0 \quad \text{u} \quad (\beta^2 + 10\beta + 29)y = 0$$

Auxiliary eq is $\beta^2 + 10\beta + 29 = 0$

$$D = \frac{-10 \pm \sqrt{100 - 4 \times 29}}{2} = \frac{-10 \pm \sqrt{-16}}{2}$$

$$= \frac{-10 \pm 4i}{2} = -5 \pm 2i$$

($\alpha = -5$, $\beta = 2$)

General soln:

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{-5x} [c_1 \cos 2x + c_2 \sin 2x]$$

Solve the following diff. eqs

$$(1) \frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + 100 \frac{dy}{dx} - 100y = 0$$

$$(2) \frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0$$

$$(3) (D^2 + D + 1)y = 0$$

$$(4) \frac{d^4 x}{dt^4} + 4x = 0$$

Answers

$$(1) (D^3 - D^2 + 100D - 100)y = 0$$

$$\text{A.E. is } D^3 - D^2 + 100D - 100 = 0$$

$$\Rightarrow D^2(D-1) + 100(D-1) = 0$$

$$\Rightarrow (D^2 + 100)(D-1) = 0$$

$$D^2 + 100 = 0 \Rightarrow D^2 = -100 \Rightarrow D = \pm 10i$$

$$\text{Also } D-1 = 0 \Rightarrow D = 1$$

Hence the roots are $D = 1, \pm 10i$

General solution is given by

$$y = c_1 e^x + e^{0x} [c_2 \cos 10x + c_3 \sin 10x] \quad (\alpha = 0, \beta = 10)$$

$$\text{u } y = c_1 e^x + c_2 \cos 10x + c_3 \sin 10x$$

$$(2) \frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0$$

$$\text{i.e. } (D^3 - 4D^2 + D + 6)y = 0$$

Auxiliary eq. is given by

$$D^3 - 4D^2 + D + 6 = 0$$

$D = -1$ is a root

$\therefore D+1$ is a factor

Dividing $D^3 - 4D^2 + D + 6$ by $D+1$

we get $(D^2 - 5D + 6)$

$$\begin{aligned} \therefore D^3 - 4D^2 + D + 6 &= (D+1)(D^2 - 5D + 6) \\ &= (D+1)(D-2)(D-3) \end{aligned}$$

$$\therefore D = 1, 2, 3$$

Hence gen. soln is given by

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$$

$$(3) (D^2 + D + 1)y = 0$$

$$\text{A.E. is } D^2 + D + 1 = 0$$

$$D = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$$

General solution is given by $(\alpha = -\frac{1}{2}, \beta = \frac{\sqrt{3}}{2})$

$$y = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{-\frac{x}{2}} \left[c_1 \cos \frac{\sqrt{3}x}{2} + c_2 \sin \frac{\sqrt{3}x}{2} \right]$$

$$(4) \frac{d^4 x}{dt^4} + 4x = 0 \Rightarrow (D^4 + 4)x = 0$$

$$\text{A.E. is given by } D^4 + 4 = 0$$

$$\text{i.e. } (D^4 + 4D^2 + 4) - 4D^2 = 0$$

$$\text{i.e. } (D^2 + 2)^2 - (2D)^2 = 0$$

$$\begin{array}{r} D^2 - 5D + 6 \\ D+1 \overline{) D^3 - 4D^2 + D + 6} \\ \underline{D^3 + D^2} \\ -5D^2 + D \\ \underline{-5D^2 - 5D} \\ 6D + 6 \\ \underline{6D + 6} \\ 0 \end{array}$$

$$i \quad (D^2 + 2D + 2)(D^2 + 2 - 2D) = 0$$

$$\Rightarrow D^2 + 2D + 2 = 0,$$

$$i \quad D = \frac{-2 \pm \sqrt{4-8}}{2}$$

$$= \frac{-2 \pm \sqrt{-4}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$(\alpha = -1, \beta = 1)$$

$$D^2 - 2D + 2 = 0$$

$$D = \frac{2 \pm \sqrt{4-8}}{2}$$

$$= \frac{2 \pm 2i}{2}$$

$$= 1 \pm i$$

$$(\alpha = 1, \beta = 1)$$

General solution:

$$y = e^{-t} [C_1 \cos t + C_2 \sin t] + e^t [C_3 \cos t + C_4 \sin t]$$

Solve the following diff. eqs.

$$(1) \frac{d^4 y}{dx^4} - y = 0$$

$$(2) (D^2 - 6D + 9)y = 0$$

$$(3) \frac{d^3 y}{dx^3} - 13 \frac{dy}{dx} + 12y = 0$$

$$(4) (D^3 + 1)y = 0$$

$$(5) \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$$

Solution of non-homogeneous diff. eq.

Consider the diff. eq.

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$i \quad (a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

$$ii \quad f(D)y = \phi(x)$$

$$\text{where } f(D) = a_0 D^2 + a_1 D + a_2$$

Homogeneous part of this eq is given by

$$(a_0 D^2 + a_1 D + a_2)y \rightarrow (2)$$

$$i \quad f(D)y = 0$$

General solution of eq (2) is called complementary function (CF) and any solution of (2) is called a particular integral (PI)

Theorem

Consider the diff. eq

$$(a_0 D^2 + a_1 D + a_2)y = \phi(x) \rightarrow (1)$$

It's homogeneous part is given by

$$(a_0 D^2 + a_1 D + a_2)y = 0 \rightarrow (2)$$

General solution of eq (1) is given by

$$y = CF + PI$$

Proof Let y_g be the general soln of eq (2)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2)y_g = 0 \rightarrow (3)$$

and y_p be a particular soln of eq (1)

$$\text{i.e. } (a_0 D^2 + a_1 D + a_2)y_p = \phi(x) \rightarrow (4)$$

$$\text{Let } y = y_g + y_p \rightarrow (5)$$

$$\begin{aligned} (a_0 D^2 + a_1 D + a_2)y &= (a_0 D^2 + a_1 D + a_2)(y_g + y_p) \\ &= (a_0 D^2 + a_1 D + a_2)y_g + (a_0 D^2 + a_1 D + a_2)y_p \\ &= 0 + \phi(x) \quad (\text{using (3) and (4)}) \\ &= \phi(x) \end{aligned}$$

This shows that $y = y_g + y_p$ satisfies eq (1)

$\therefore$ It is a solution of eq (1).

Since y_g is the gen. soln of eq (2)

it contains two arbitrary constants

eq (3) gives the gen. soln of (1).

$$\text{i.e. } y = y_g + y_p = CF + PI \text{ is the gen. soln.}$$

Special methods of finding Particular integral

consider the diff. eq

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = \phi(x)$$

$$\text{i.e. } f(D)y = \phi(x) \quad \text{where } f(D) = a_0 D^2 + a_1 D + a_2 \quad \text{--- (1)}$$

It's gen. soln is given by

$$y = CF + PI$$

case (I) $\phi(x) = e^{ax}$

$$D(e^{ax}) = a e^{ax}, \quad D^2(e^{ax}) = a^2 e^{ax}$$

$$\begin{aligned} \therefore f(D)(e^{ax}) &= (a_0 D^2 + a_1 D + a_2) e^{ax} \\ &= a_0 a^2 e^{ax} + a_1 a e^{ax} + a_2 e^{ax} \\ &= (a_0 a^2 + a_1 a + a_2) e^{ax} \\ &= f(a) e^{ax} \end{aligned}$$

Operating both sides by $f(D)^{-1} = \frac{1}{f(D)}$, we get

$$\frac{1}{f(D)} \cdot f(D) e^{ax} = \frac{1}{f(D)} f(a) e^{ax}$$

$$\text{i.e. } e^{ax} = f(a) \cdot \frac{1}{f(D)} e^{ax}$$

$$\therefore \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$$

provided $f(a) \neq 0$.

$$\text{i.e. } PI = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad \text{if } f(a) \neq 0$$

Note $f(a) = 0 \Rightarrow a$ is a root of the A.E.
 $f(D) = 0$

If 'a' is a root of multiplicity r then

$$PI = \frac{x^r}{f^{(r)}(a)} e^{ax} \quad \text{where } f^{(r)}(a) = f^{(r)}(D) \text{ at } D=a$$

$$\text{when } r=1, \quad PI = \frac{x}{f'(a)} e^{ax}$$

$$\text{when } r=2 \quad PI = \frac{x^2}{f''(a)} e^{ax} \dots$$

Examples

(1) Solve $\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = e^{-2x}$

Associated eq is given by

$$\frac{d^2 y}{dx^2} - 13 \frac{dy}{dx} + 12y = 0 \Rightarrow (D^2 - 13D + 12)y = 0$$

A.E. is given by $D^2 - 13D + 12 = 0$

$$\Rightarrow (D-1)(D-12) = 0 \Rightarrow D = 1, 12$$

$$\therefore CF = C_1 e^x + C_2 e^{12x}$$

$$PI = \frac{1}{f(D)} e^{-2x} = \frac{1}{D^2 - 13D + 12} e^{-2x} \quad (a = -2)$$

$$= \frac{1}{(-2)^2 - 13(-2) + 12} e^{-2x} \quad \left(\begin{array}{l} \text{Replacing} \\ D \text{ by } -2 \end{array} \right)$$

$$= \frac{1}{4 + 26 + 12} e^{-2x}$$

$$= \frac{1}{42} e^{-2x}$$

Gen. soln. is given by

$$y = CF + PI = C_1 e^x + C_2 e^{12x} + \frac{1}{42} e^{-2x}$$

(2) Solve $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = e^{2x}$

$$\Rightarrow (D^2 + 3D - 10)y = e^{2x}$$

A.E. is $D^2 + 3D - 10 = 0$

$$\Rightarrow (D-2)(D+5) = 0 \Rightarrow D = 2, -5$$

$$CF = C_1 e^{2x} + C_2 e^{-5x}$$

$$PI = \frac{1}{f(D)} e^{2x} = \frac{1}{D^2 + 3D - 10} e^{2x} \quad \left(\begin{array}{l} a=2 \text{ which is} \\ \text{single root of A.E.} \end{array} \right)$$

$$= \frac{x}{f'(a)} e^{2x}$$

$$= \frac{x}{7} e^{2x}$$

$$\left\{ \begin{array}{l} f(D) = D^2 + 3D - 10 \\ f'(D) = 2D + 3 \\ f'(a) = 2(2) + 3 = 7 \end{array} \right.$$

gen. soln. $y = CF + PI = C_1 e^{2x} + C_2 e^{-5x} + \frac{x}{7} e^{2x}$

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$$3) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 36y = e^{6x}$$

$$\text{or } (D^2 - 12D + 36)y = e^{6x}$$

$$\text{A.E. is } D^2 - 12D + 36 = 0$$

$$\text{or } (D-6)^2 = 0 \Rightarrow D = 6, 6$$

$$\text{CF} = (C_1 + C_2 x) e^{6x}$$

$$\text{PI} = \frac{1}{D^2 - 12D + 36} e^{6x}$$

$$= \frac{x^2}{f''(a)} e^{6x}$$

$$= \frac{x^2}{2} e^{6x}$$

($a=6$ which is a double root of the A.E.)

$$\left| \begin{array}{l} f(D) = D^2 - 12D + 36 \\ f'(D) = 2D - 12 \\ f''(D) = 2 \therefore f''(6) = 2 \end{array} \right|$$

gen. soln is $y = \text{CF} + \text{PI}$

$$= (C_1 + C_2 x) e^{6x} + \frac{x^2}{2} e^{6x}$$

Solve (1) $\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y = 2e^x + 10e^{-5x}$

(2) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 4y = 4e^x - 18e^{-x}$

(3) $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x + 4e^{3x}$

Answers

(1) $(D^2 + 6D + 5)y = 2e^x + 10e^{-5x}$

$$\text{A.E. is } D^2 + 6D + 5 = 0$$

$$\text{or } (D+1)(D+5) = 0 \Rightarrow D = -1, -5$$

$$\text{CF} = C_1 e^{-x} + C_2 e^{-5x}$$

$$\text{PI} = \frac{1}{D^2 + 6D + 5} (2e^x + 10e^{-5x})$$

$$= \frac{1}{D^2 + 6D + 5} (2e^x) + \frac{1}{D^2 + 6D + 5} (10e^{-5x})$$

$$= \text{PI}_1 + \text{PI}_2$$

$$\text{PI}_1 = \frac{1}{D^2 + 6D + 5} (2e^x) = \frac{1}{1^2 + 6(1) + 5} (2e^x) \quad (a=1 \text{ which is not a root of A.E.})$$

$$PI_1 = \frac{1}{12} (2e^x) = \frac{1}{6} e^x$$

$$PI_2 = \frac{1}{D^2 + bD + 5} (10e^{-5x})$$

$$= \frac{x}{f'(a)} (10e^{-5x})$$

$$= \frac{x}{-4} (10e^{-5x})$$

$$= -\frac{5x}{2} e^{-5x}$$

($a = -5$ which is a single root of the A.E.)

$$f(D) = D^2 + bD + 5$$

$$f'(D) = 2D + b$$

$$f'(a) = 2(-5) + b = -4$$

gen. soln $y = CF + PI$

$$y = CF + PI_1 + PI_2$$

$$\therefore y = c_1 e^{-x} + c_2 e^{-5x} + \frac{1}{6} e^x - \frac{5x}{2} e^{-5x}$$

$$(2) (D^2 - 3D + 4)y = 4e^x - 18e^{-x}$$

A.E. is $D^2 - 3D + 4 = 0$

$$D = \frac{3 \pm \sqrt{9 - 16}}{2} = \frac{3 \pm i\sqrt{7}}{2}$$

$$(\alpha = \frac{3}{2}, \beta = \frac{\sqrt{7}}{2})$$

$$CF = e^{\alpha x} [c_1 \cos \beta x + c_2 \sin \beta x]$$

$$= e^{\frac{3x}{2}} [c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x]$$

$$PI = \frac{1}{D^2 - 3D + 4} (4e^x - 18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{D^2 - 3D + 4} (-18e^{-x})$$

$$= \frac{1}{D^2 - 3D + 4} (4e^x) + \frac{1}{(-1)^2 - 3(-1) + 4} (-18e^{-x})$$

$$= \frac{1}{2} (4e^x) + \frac{1}{8} (-18e^{-x})$$

$$= 2e^x - \frac{9}{4} e^{-x}$$

gen. soln $y = CF + PI$

$$= e^{\frac{3x}{2}} [c_1 \cos \frac{\sqrt{7}}{2} x + c_2 \sin \frac{\sqrt{7}}{2} x] + 2e^x - \frac{9}{4} e^{-x}$$

$$(3) (D^2 - 3D + 2)y = e^x + 4e^{3x}$$

$$\text{A.E. is } D^2 - 3D + 2 = 0 \Rightarrow (D-1)(D-2) = 0$$

$$\therefore D = 1, 2$$

$$CF = c_1 e^x + c_2 e^{2x}$$

$$PI = \frac{1}{D^2 - 3D + 2} (e^x + 4e^{3x})$$

$$= \frac{1}{D^2 - 3D + 2} (e^x) + \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 - 3D + 2} (e^x)$$

(Here $a=1$ which is a single root of the A.E.)

$$= \frac{x}{f'(a)} e^{ax}$$

$$= \frac{x}{-1} e^x = -xe^x$$

$$\left(\begin{array}{l} f(D) = D^2 - 3D + 2 \\ f'(D) = 2D - 3 \\ f'(a) = 2(1) - 3 = -1 \end{array} \right)$$

$$PI_2 = \frac{1}{D^2 - 3D + 2} (4e^{3x})$$

(Here $a=3$ which is not a root of the A.E.)

$$= \frac{1}{3^2 - 3(3) + 2} (4e^{3x}) = \frac{1}{2} (4e^{3x}) = 2e^{3x}$$

$$\text{Gen. Soln } y = CF + PI = c_1 e^x + c_2 e^{2x} - xe^x + 2e^{3x}$$

Solve the following diff. eqs

$$(1) (D^2 - 5D + 6)y = e^{5x}$$

$$(2) (D^2 + 5D + 7)y = e^{-x}$$

$$(3) (D^2 + 4D + 4)y = e^{-3x}$$

$$(4) \frac{d^2 y}{dx^2} - 9y = 5e^{4x} + 2e^{3x}$$

$$(5) (D^2 - 6D + 8)y = (e^{2x} - 1)^2$$